

# Big Data

Whenever we think of big data, words like predicting preferences, behaviors, psychology, and audience come to mind. So today let us understand how big data breaks into our lives.

First, we need to know how it works. Big data uses concepts from mathematical analysis, optimization, inductive statistics, and identification of nonlinear systems to infer patterns (regression, nonlinear relationships, and causal effects) from large data sets with low information density, revealing relationships and dependencies, or predicting outcomes and behaviors. In this way it observes our life in any part and stores this information as a base, then we get to know what would we like most?

There are still a lot of important properties in Big Data.

## Velocity

The speed at which data is generated and processed to meet the needs and challenges of growth and development. Big data is often generated in real time. Compared to small data, big data is generated more continuously. Two velocities associated with big data are the frequency of generation and the frequency of processing, recording, and publishing.

## Authenticity

The authenticity or reliability of the data, i.e., the quality and value of the data. Big data must not only be large, but also reliable to realize value in analysis. The quality of the data captured can vary greatly, affecting accurate analysis.

## Value

The value of information that can be realized by processing and analyzing large data sets. Value can also be measured by evaluating other qualities of big data. Value may also represent the profitability of information retrieved from big data analysis.

Second, technology is the base of any Science. Developed economies are increasingly using data-intensive technologies. There are 4.6 billion mobile phone users worldwide, and between 1 billion and 2 billion people access the Internet. Between 1990 and 2005, more than 1 billion people entered the middle class, meaning more people became more literate, which in turn led to information growth. In 1986, the world's effective capacity for information exchanged over telecommunications networks was 281 petabytes, 471 petabytes in 1993, 2.2

exabytes in 2000, and 65 exabytes in 2007. Internet traffic is predicted to reach 667 exabytes per year by 2014. According to one estimate, one-third of the world's stored information is in the form of alphanumeric text and still image data, the formats most useful for most big data applications. This also shows the potential of untapped data, namely in the form of video and audio content.

Thirdly, everything is valuable only when it can make some value. So applications are the key to make things useful. Walmart processes more than 1 million customer transactions per hour, which are imported into a database estimated to contain more than 2.5 petabytes (2,560 terabytes) of data—167 times the information contained in all the books in the Library of Congress.

Windermere Real Estate uses location information from nearly 100 million drivers to help new home buyers determine their typical driving time to and from work at different times of the day.

The decoding of the human genome originally took 10 years; it can now be done in less than a day. Over the past decade, DNA sequencers have increased the cost of sequencing by 10,000, a factor of 100 less than the cost reduction predicted by Moore's Law.

And also, there are three important factors in using big data in marketing:

1. Big data provides marketers with the ability to discover patterns in customer behavior, as all human behavior is quantified into readable numbers for marketers to analyze and study. In addition, big data can also be viewed as a customized product recommendation tool. Specifically, since big data is effective in analyzing customer purchasing behavior and browsing patterns, this technology can help companies promote specific personalized products to specific customers.
2. Real-time market responsiveness is important for marketers because they are able to shift marketing efforts and correct current trends, which helps to maintain relevance to consumers. This can provide businesses with the necessary information to predict consumer desires and needs in advance.
3. Big data is driving data-driven marketing efforts. New models and algorithms are being developed to make significant predictions about certain economic and social conditions.

Finally, Big Data is the production of large population and explosion of information. It is based on the developing technology and increasing demand in consumption of target.

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